

# fn sample\_geometric\_exp\_fast

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Proves soundness of `fn sample_geometric_exp_fast` in `mod.rs` at commit `0be3ab3e6` (outdated<sup>1</sup>). This proof is an adaptation of subsection 5.2 of [CKS20].

## 1 Hoare Triple

### Precondition

#### Compiler-verified

- Argument `x` is of type `RBig`, a bignum rational.

#### Caller-verified

- $x \geq 0$

### Pseudocode

```
1 def sample_geometric_exp_fast(x: RBig) -> int:
2     if x == 0:
3         return 0
4
5     s, t = x.into_parts()
6
7     while True:
8         u = sample_uniform_ubig_below(t) #
9         d = sample_bernoulli_exp(Rational(u, t)) #
10        if d:
11            break
12
13    v = sample_geometric_exp_slow(1) #
14    z = u + t * v
15    return z // s
```

### Postcondition

**Theorem 1.1.** For any setting of the input parameter `x` such that the given preconditions hold, `sample_geometric_exp_fast` either returns `Err(e)` due to a lack of system entropy, or `Ok(out)`, where `out` is distributed as  $\text{Geometric}(1 - \exp(-x))$ .

Assume the preconditions are met.

**Lemma 1.2.** There exist a non-negative integer  $s$  and positive integer  $t$  such that  $x = s/t$ . If  $x > 0$ , then  $s \geq 1$ .

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<sup>1</sup>See new changes with `git diff 0be3ab3e6..9cbde10f rust/src/traits/samplers/cks20/mod.rs`

*Proof.* Since  $x$  is of type **Rbig** and  $x \geq 0$ , it can be written as  $x = s/t$  for some non-negative integer  $s$  and positive integer  $t$ . This is why **RBig.into\_parts** is infallible. If  $x > 0$ , then its numerator is positive, so  $s \geq 1$ .  $\square$

**Lemma 1.3.** **sample\_geometric\_exp\_fast** only returns **Err(e)** when there is a lack of system entropy.

*Proof.* If  $x = 0$ , the function returns **Ok(0)** immediately, so it cannot return **Err(e)**. Assume  $x > 0$ . By Lemma 1.2, there exist integers  $s \geq 1$  and  $t > 0$  with  $x = s/t$ . Since  $t$  is a positive integer, the preconditions on **sample\_uniform\_ubig\_below** are met. Any returned value  $u$  therefore lies in  $\{0, 1, \dots, t-1\}$ , so  $u/t \geq 0$ , and the preconditions on **sample\_bernoulli\_exp** are also met. The input on line 13 is the non-negative rational 1, so the preconditions on **sample\_geometric\_exp\_slow** are met as well. By their definitions, each of these fallible calls can only return an error due to a lack of system entropy. These are the only fallible calls in **sample\_geometric\_exp\_fast**, so the function only returns **Err(e)** when there is a lack of system entropy.  $\square$

We now establish some lemmas that will be useful in proving the distribution of **out**.

**Lemma 1.4.** Suppose  $x > 0$ . Conditioned on the loop on lines 8–9 terminating without returning **Err(e)**, the value  $u$  at loop exit is a realization of a random variable  $U_{\text{acc}}$  supported on  $\{0, 1, \dots, t-1\}$ , where

$$P[U_{\text{acc}} = u] = \frac{1 - e^{-1/t}}{1 - e^{-1}} e^{-u/t}.$$

*Proof.* By Lemma 1.2,  $t > 0$ , so in each loop iteration the call on line 8 samples a random variable  $U \sim \text{Uniform}(0, t)$  supported on  $\{0, 1, \dots, t-1\}$ . Conditioned on a realization  $u$  of  $U$ , the call on line 9 samples  $D \sim \text{Bernoulli}(\exp(-u/t))$  by the specification of **sample\_bernoulli\_exp**.

Each iteration uses fresh randomness, and the loop stops at the first iteration for which  $D = \top$ . Therefore the value returned by the loop has the same distribution as  $U$  conditioned on  $D = \top$ . For any  $u \in \{0, 1, \dots, t-1\}$ ,

$$\begin{aligned} P[U_{\text{acc}} = u] &= P[U = u \mid D = \top] \\ &= \frac{P[U = u]P[D = \top \mid U = u]}{P[D = \top]} && \text{by Bayes' theorem} \\ &= \frac{(1/t)e^{-u/t}}{\frac{1}{t} \sum_{k=0}^{t-1} e^{-k/t}} \\ &= \frac{(1/t)e^{-u/t}}{\frac{1}{t} \frac{1 - e^{-1}}{1 - e^{-1/t}}} \\ &= \frac{1 - e^{-1/t}}{1 - e^{-1}} e^{-u/t}. \end{aligned}$$

$\square$

**Lemma 1.5.** Suppose  $x > 0$  and **sample\_geometric\_exp\_fast** returns **Ok(out)**. Then the value  $z$  is a realization of a random variable  $Z \sim \text{Geometric}(1 - \exp(-1/t))$ .

*Proof.* By Lemma 1.4, the value  $u$  exiting the loop is distributed as  $U_{\text{acc}}$ . The call on line 13 returns a value  $v$  distributed as  $V \sim \text{Geometric}(1 - \exp(-1))$  by the specification of **sample\_geometric\_exp\_slow**. Since line 13 is executed after the loop and uses fresh randomness,  $U_{\text{acc}}$  and  $V$  are independent.

For any  $z$ , define  $u_z := z \bmod t$  and  $v_z := \lfloor z/t \rfloor$ , so that  $z = u_z + t \cdot v_z$ . Then

$$\begin{aligned}
P[Z = z] &= P[U_{\text{acc}} = u_z, V = v_z] && \text{since } z = u_z + t \cdot v_z \\
&= P[U_{\text{acc}} = u_z]P[V = v_z] && \text{by independence} \\
&= \frac{1 - e^{-1/t}}{1 - e^{-1}} e^{-u_z/t} \cdot (1 - e^{-1}) e^{-v_z} && \text{by Lemma 1.4} \\
&= (1 - e^{-1/t}) e^{-(u_z/t + v_z)} \\
&= (1 - e^{-1/t}) e^{-z/t}.
\end{aligned}$$

Therefore  $Z \sim \text{Geometric}(1 - \exp(-1/t))$ .  $\square$

**Lemma 1.6.** [CKS20] Fix  $p \in (0, 1]$ . Let  $G$  be a  $\text{Geometric}(1 - p)$  random variable, and  $n \geq 1$  be an integer. Then  $\lfloor G/n \rfloor$  is a  $\text{Geometric}(1 - q)$  random variable with  $q = p^n$ .

*Proof.*

$$\begin{aligned}
P[\lfloor G/n \rfloor = k] &= P[nk \leq G < (k+1)n] && \text{any } G \text{ in the interval maps to } k \\
&= \sum_{l=kn}^{(k+1)n-1} (1-p)p^l \\
&= (1-p^n)p^{nk} \\
&= (1-q)q^k
\end{aligned}$$

$\square$

**Lemma 1.7.** If the outcome of `sample_geometric_exp_fast` is `Ok(out)`, then `out` is distributed as  $\text{Geometric}(1 - \exp(-x))$ .

*Proof.* If  $x = 0$ , the function returns `Ok(0)` immediately. Since  $\text{Geometric}(1 - \exp(-0)) = \text{Geometric}(0)$  is the point mass at 0, the claim holds.

Assume  $x > 0$ . By Lemma 1.2, there exist integers  $s \geq 1$  and  $t > 0$  with  $x = s/t$ . By Lemma 1.5, `z` is a realization of  $Z \sim \text{Geometric}(1 - \exp(-1/t))$ . Applying Lemma 1.6 with  $p = \exp(-1/t)$  and  $n = s$ , we conclude that  $\lfloor Z/s \rfloor$  is distributed as  $\text{Geometric}(1 - \exp(-s/t))$ . Finally, `out` =  $\lfloor Z/s \rfloor$  and  $s/t = x$ , so `out` is distributed as  $\text{Geometric}(1 - \exp(-x))$ .  $\square$

*Proof.* Theorem 1.1 holds by 1.3 and 1.7.  $\square$

## References

[CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.